

13.5 False discovery rate control

With hundreds of hypotheses, as when a regression screens many candidate predictors, family-wise control becomes self-defeating: the threshold α/m shrinks with m , and power disappears exactly when there is most to find. The false discovery rate of Definition 13.1.1 asks instead that, on average, only a small *proportion* of the rejections be false. This section proves that the procedure of Benjamini and Hochberg controls it for independent p -values, and describes what is known under dependence.

13.5.1 The Benjamini–Hochberg procedure

As before, $p_{(1)} \leq \dots \leq p_{(m)}$ are the ordered p -values, $H_{(i)}$ the corresponding hypotheses, and $q \in (0, 1)$ is the target false discovery rate.

Algorithm.

Benjamini–Hochberg (BH) procedure at level q . Let

$$\hat{k} = \max \left\{ k \in \{1, \dots, m\} : p_{(k)} \leq \frac{kq}{m} \right\},$$

and reject $H_{(1)}, \dots, H_{(\hat{k})}$. If no k qualifies, reject nothing.

The thresholds rise linearly from Bonferroni's q/m to q . The procedure is a **step-up** procedure: it rejects everything up to the *last* p -value on or below the line, including smaller p -values that lie above the line at their own rank (Figure 13.5.1).

Why is the line the right one? Suppose k hypotheses are rejected, all with p -values at most $t = kq/m$. The m_0 true hypotheses have uniform p -values, so about $m_0 t \leq mt$ of them fall below t by chance, and the proportion of false ones among the k rejections is about $mt/k = q$ at most. The BH procedure chooses the largest threshold at which this estimate of the false discovery proportion is at most q . The theorem makes the heuristic exact.

Two simple descriptions of the rejected set are useful. Write $N(t) = \#\{j : p_j \leq t\}$ for the number of p -values at most t . Since $p_{(k)} \leq t$ iff $N(t) \geq k$,

$$\hat{k} = \max \left\{ k : N(kq/m) \geq k \right\}. \quad (13.5.1)$$

And the rejected hypotheses are exactly those with $p_j \leq \hat{k}q/m$. Indeed, every rejected p -value is at most $p_{(\hat{k})} \leq \hat{k}q/m$. Conversely, if some p_j with j not rejected were at most $\hat{k}q/m$, then $N(\hat{k}q/m) \geq \hat{k} + 1$, hence $N((\hat{k} + 1)q/m) \geq \hat{k} + 1$, and $\hat{k} + 1$ would qualify, contradicting the maximality of $\hat{k}$.

13.5.2 Control under independence

Lemma 13.5.1. Replacing one p -value by zero

Fix i , and let $\hat{k}^{(i)}$ be the number of rejections of the BH procedure applied to the p -values

with p_i replaced by 0. Then $\hat{k}^{(i)}$ is a function of $(p_j)_{j \neq i}$ alone, and for each $k = 1, \dots, m$,

$$\{H_i \text{ is rejected and } \hat{k} = k\} = \{p_i \leq kq/m \text{ and } \hat{k}^{(i)} = k\}.$$

Proof. The first claim is clear. Let $N^{(i)}$ count the modified p -values. Then $N^{(i)}(t) = N(t)$ for $t \geq p_i$, because at such t the p -value p_i is counted in both. By the second description above, H_i is rejected and $\hat{k} = k$ iff $p_i \leq kq/m$ and $\hat{k} = k$. Assume $p_i \leq kq/m$. By [Eq. 13.5.1](#), $\hat{k} = k$ iff $N(kq/m) \geq k$ and $N(k'q/m) < k'$ for every $k' > k$. These conditions involve N only at points $t \geq kq/m \geq p_i$, where N and $N^{(i)}$ agree, so they hold iff the same conditions hold for $N^{(i)}$, that is, iff $\hat{k}^{(i)} = k$. ■

Theorem 13.5.2. Benjamini–Hochberg

Suppose that for each true hypothesis H_i , $i \in I_0$, the p -value p_i is independent of the other $m - 1$ p -values and satisfies $\Pr(p_i \leq u) \leq u$ for $0 \leq u \leq 1$. Then the BH procedure at level q has

$$\text{FDR} \leq \frac{m_0}{m} q \leq q,$$

with equality in the first inequality when every p_i , $i \in I_0$, is uniform on $(0, 1)$.

Proof. The false discovery proportion is $Q = \sum_{i \in I_0} 1_{\{H_i \text{ rejected}\}} / \max(\hat{k}, 1)$. A rejection forces $\hat{k} \geq 1$, so

$$\text{FDR} = \sum_{i \in I_0} \sum_{k=1}^m \frac{1}{k} \Pr(H_i \text{ rejected}, \hat{k} = k) = \sum_{i \in I_0} \sum_{k=1}^m \frac{1}{k} \Pr\left(p_i \leq \frac{kq}{m}\right) \Pr(\hat{k}^{(i)} = k),$$

by [Lemma 13.5.1](#) and the independence of p_i from $\hat{k}^{(i)}$. Now $\Pr(p_i \leq kq/m) \leq kq/m$, so the inner sum is at most $(q/m) \sum_{k=1}^m \Pr(\hat{k}^{(i)} = k) = q/m$, because $\hat{k}^{(i)} \geq 1$ always (the replaced p -value 0 lies below the first threshold). Summing over the m_0 true hypotheses gives $\text{FDR} \leq m_0 q/m$. With uniform null p -values every inequality is an equality. ■

The false hypotheses' p -values may depend on each other in any way. With uniform null p -values the false discovery rate is exactly $\pi_0 q$, $\pi_0 = m_0/m$. With $m = 100$ independent z -tests, 90 of them null and the rest shifted by 3, 20000 simulated repetitions estimate it at 0.0439, against the exact 0.0450.

When all hypotheses are true the false discovery rate is the familywise error rate (Proposition 13.1.1), so BH is a level- q test of the global null. It rejects everything Bonferroni's and Holm's procedures at level q reject ([Exercise \(B2\)](#)). But it does *not* control the familywise error rate strongly: once many false hypotheses are rejected, the thresholds for the rest approach q .

13.5.3 Dependent tests

In a regression the t statistics share s , and their numerators are correlated unless $\mathbf{X}$ has orthogonal columns. Two results, stated here without proof, cover much of what is needed.

Positive dependence. Benjamini and Yekutieli (2001) showed that $\text{FDR} \leq m_0 q/m$ remains true if the p -values are *positively regression dependent on each one from the subset of true hypotheses* (PRDS): for every true H_i and every increasing set D of p -value vectors, $\Pr((p_1, \dots, p_m) \in D \mid p_i = u)$ is nondecreasing in u . One-sided tests based on jointly normal statistics with nonnegative correlations are an example; two-sided tests of correlated statistics need not be.

Arbitrary dependence. They also showed that under *any* dependence, BH at level q/c_m with $c_m = \sum_{i=1}^m 1/i \approx \log m + 0.577$ has false discovery rate at most $m_0 q/m$ (Exercise (C1)). For $m = 100$, $c_m = 5.187$, so a target of 0.05 becomes an effective level of 0.0096.

BH is routinely applied to regression t tests, and simulations, including the one below, find its false discovery rate below q . That is evidence, not a theorem; when a guarantee is essential, the Benjamini–Yekutieli version is the safe choice.

13.5.4 Adjusted p -values and q -values

As for Holm's procedure, the BH decisions for all levels at once are summarized by **adjusted p -values**

$$\tilde{p}_{(i)} = \min_{j \geq i} \min \left\{ 1, \frac{m p_{(j)}}{j} \right\}, \quad (13.5.2)$$

with $H_{(i)}$ rejected at level q iff $\tilde{p}_{(i)} \leq q$ (Exercise (B1)). Since BH controls the rate at $\pi_0 q$, it is conservative when many hypotheses are false. **Adaptive** procedures estimate π_0 , for instance by twice the proportion of p -values above one half, and run BH at level $q/\hat{\pi}_0$. Storey (2002) developed this approach and defined the **q -value** of a hypothesis, the smallest false discovery rate at which it would be rejected; with $\hat{\pi}_0 = 1$ the q -values are the adjusted values Eq. 13.5.2. The gain is largest when π_0 is well below one.

13.5.5 Screening regression coefficients

Example. One hundred candidate regressors

A simulated regression has $n = 250$ observations, an intercept and 100 regressors with independent standard normal entries, leaving $\nu = 149$ error degrees of freedom. The first 10 slopes are nonzero, each chosen so that the noncentrality of its t statistic is 3; the other 90 are zero. The 100 two-sided t tests are run on one data set at level 0.05:

procedure	rejections	of which false
unadjusted, $p \leq 0.05$	9	2
Bonferroni	1	0
Holm	1	0
BH, $q = 0.05$	7	1

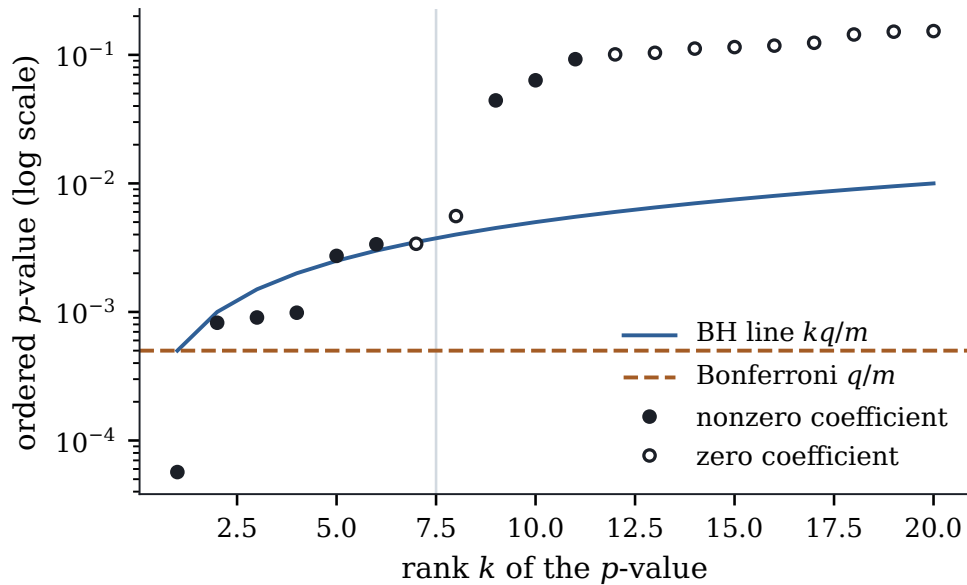


Figure 13.5.1: The twenty smallest of 100 ordered p -values (log scale), with the BH line kq/m and Bonferroni's threshold q/m , $q = 0.05$. Filled points are nonzero coefficients; BH rejects the seven left of the vertical line.

Bonferroni's threshold is $0.05/100 = 0.0005$, and only one p -value is below it. The BH line reaches $7 \times 0.05/100 = 0.0035$ at rank 7, and the seventh smallest p -value is 0.0034 (Figure 13.5.1). Six of the seven BH discoveries are real.

```
import numpy as np
from scipy import stats

q = 0.05

rng = np.random.default_rng(20260919)
n, m = 250, 100
X = np.column_stack([np.ones(n), rng.normal(size=(n, m))])
Qx, Rx = np.linalg.qr(X)
nu = n - m - 1
c_diag = np.diag(np.linalg.inv(Rx.T @ Rx))[1:]
# 149 error degrees of freedom
# [(X'X)^{-1}]_{jj} for the slopes

def t_statistics(Y):
    """t statistics of the 100 slopes for each column of Y."""
    coef = np.linalg.solve(Rx, Qx.T @ Y)
    resid = Y - X @ coef
    s2 = np.sum(resid ** 2, axis=0) / nu
    return coef[1:] / np.sqrt(np.outer(c_diag, s2))
```

[Run this code online](#)

```
def bh(p, q):
    """Benjamini-Hochberg step-up at level q; returns a boolean rejection vector."""
    m = len(p)
    order = np.argsort(p)
    below = np.nonzero(p[order] <= q * np.arange(1, m + 1) / m)[0]
    reject = np.zeros(m, dtype=bool)
    if below.size:
```

```

    reject[order[: below[-1] + 1]] = True
    return reject

def holm(p, alpha):
    m = len(p)
    order = np.argsort(p)
    ok = np.cumprod(p[order] <= alpha / (m - np.arange(m))).astype(bool)
    reject = np.zeros(m, dtype=bool)
    reject[order] = ok
    return reject

procedures = {
    "unadjusted": lambda p: p <= q,
    "Bonferroni": lambda p: p <= q / m,
    "Holm": lambda p: holm(p, q),
    "BH": lambda p: bh(p, q),
}

```

[Run this code online](#)

```

m1 = 10                                # the first 10 slopes are
    nonzero
beta = np.zeros(m + 1)
beta[1:m1 + 1] = 3.0 * np.sqrt(c_diag[:m1])    # noncentrality 3 when sigma =
    1
y = X @ beta + rng.normal(size=n)
p = 2 * stats.t.sf(np.abs(t_statistics(y[:, None])[:, 0]), nu)
for name, rule in procedures.items():
    rej = rule(p)
    print(f"{name:10s} rejects {rej.sum():3d}, of which {rej[m1:].sum()} are false
discoveries")

```

[Run this code online](#)

A simulation with the same design, $m_1 = 0, 5, 10, 25, 50$ nonzero slopes and 2000 responses for each, shows the average behaviour (Figure 13.5.2). With $m_1 = 10$, the false discovery rate is 0.327 without adjustment, 0.014 for Bonferroni and Holm and 0.046 for BH, and the average power is 0.845, 0.289, 0.291 and 0.476. With $m_1 = 50$, BH's power rises to 0.731 and its false discovery rate falls to 0.024, in line with $\pi_0 q$, while Bonferroni's and Holm's power stay near 0.3. BH pays with familywise errors: at least one of its rejections is false in 0.238 of the data sets with $m_1 = 10$ and 0.570 with $m_1 = 50$. With all slopes zero, its false discovery rate and familywise error rate coincide, at 0.050.

For the 21 education comparisons of Example (Holm for the education pairs), BH at $q = 0.05$ rejects 13 hypotheses: the thirteenth smallest p -value is $0.0303 \leq 13 \times 0.05/21 = 0.0310$, and no later one comes back below the line. The next section discusses how to choose between the two error rates.

13.5.6 Exercises

13.5.6.1 A. Check your understanding

Exercise. A1

Apply the BH procedure at $q = 0.05$ to the five p -values of Exercise (A1), and compute the

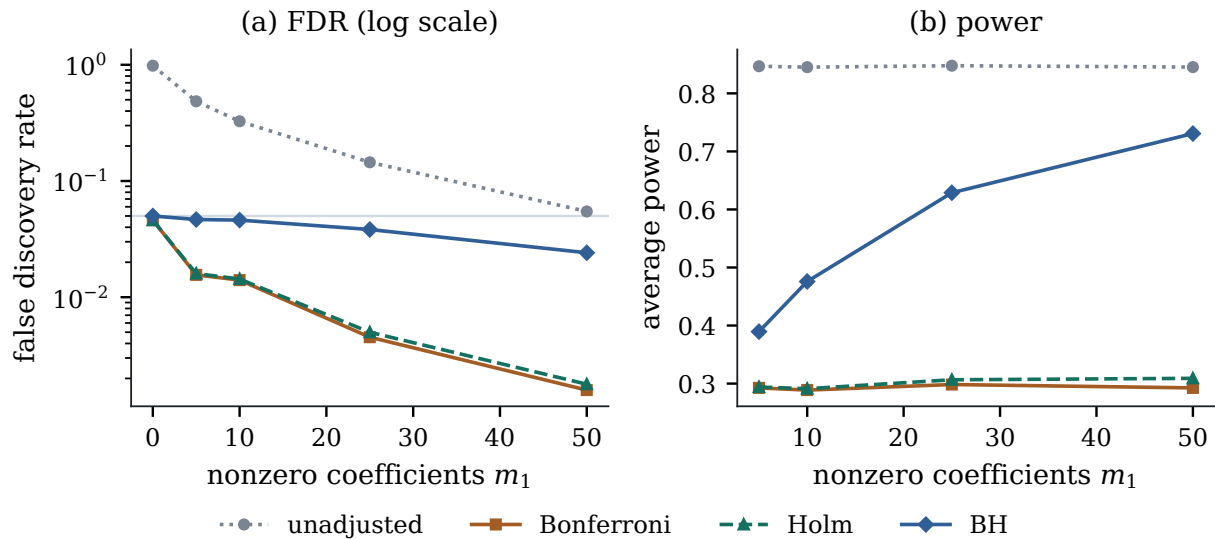


Figure 13.5.2: False discovery rate (a, log scale) and average power (b) of four procedures when m_1 of 100 regression slopes are nonzero, 2000 simulated data sets per point, nominal level 0.05. The pale line in (a) marks 0.05.

adjusted p -values [Eq. 13.5.2](#).

Solution. The thresholds are 0.01, 0.02, 0.03, 0.04, 0.05. From the top: $0.2 > 0.05$, but $0.04 \leq 0.04$, so $\hat{k} = 4$ and the first four hypotheses are rejected, one more than Holm's procedure rejects. The adjusted values are $\min(5 \times 0.001, \dots) = 0.005$, then $\min(5 \times 0.011/2, 5 \times 0.012/3, 5 \times 0.04/4, 1) = 0.02, 0.02, 0.05$ and 0.2 .

Exercise. A2

A report says: “Using BH at $q = 0.1$ we made 40 discoveries, so at most 4 of them are false.” What is wrong with this statement? What would be a correct one?

13.5.6.2 B. Practice

Exercise. B1

Prove that BH at level q rejects $H_{(i)}$ iff the adjusted value [Eq. 13.5.2](#) satisfies $\tilde{p}_{(i)} \leq q$.

Exercise. B2

Show that the BH procedure at level q rejects every hypothesis that Holm's procedure at level q rejects. *Hint:* $q/(m - i + 1) \leq iq/m$ for $1 \leq i \leq m$.

Solution. The hint holds because $i(m - i + 1) \geq m$ for $1 \leq i \leq m$: the product of two positive integers summing to $m + 1$ is at least m . If Holm rejects $H_{(1)}, \dots, H_{(i)}$, then $p_{(i)} \leq q/(m - i + 1) \leq iq/m$, so $\hat{k} \geq i$.

Exercise. B3

With $m_0 = 2$ true hypotheses with independent uniform p -values and $m - 2$ false hypotheses whose p -values are 0, find the familywise error rate of BH at level q . Compare it with the false discovery rate given by [Theorem 13.5.2](#).

13.5.6.3 C. Going deeper**Exercise. C1**

Prove the Benjamini–Yekutieli bound: under arbitrary dependence, BH at level q/c_m has false discovery rate at most $m_0 q/m$. *Hint:* write the false discovery rate as $\sum_{i \in I_0} \sum_k \Pr(p_i \in (\frac{(j-1)q'}{m}, \frac{j q'}{m}], \hat{k} = k)/k$ summed over $j \leq k$, with $q' = q/c_m$, and exchange the order of summation.